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Charged Particle Beam Device

Abstract

Even when the amount of overlay deviation between patterns located in different layers is large, correct measurement of the amount of overlay deviation is stably performed. The charged particle beam device includes a charged particle beam irradiation unit that irradiates a sample with a charged particle beam, a first detection unit that detects secondary electrons from the sample, a second detection unit that detects backscattered electrons from the sample, and an image processing unit that generates a first image including an image of a first pattern located on the surface of the sample based on an output of the first detection unit, and generates a second image including an image of a second pattern located in a lower layer than the surface of the sample based on an output of the second detection unit. A control unit adjusts the position of a measurement area in the first image based on a first template image for the first image, and adjusts the position of a measurement area in the second image based on a second template image for the second image.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] The present application is a Continuation of U.S. patent application Ser. No. 18/333,637, filed Jun. 13, 2023 which is a Continuation of U.S. patent application Ser. No. 17/518,739, filed Nov. 4, 2021, now U.S. Pat. No. 11, 728, 127, issued Aug. 15, 2023, which is a Continuation of U.S. patent application Ser. No. 16/851,749, filed Apr. 17, 2020, now U.S. Pat. No. 11,195,690, issued Dec. 7, 2021, which claims priority from Japanese Patent Application No. 2019-090347, filed on May 13, 2019, the entire disclosures of which are hereby incorporated by reference into this application.

TECHNICAL FIELD

[0002] The present invention relates to a charged particle beam device.

BACKGROUND ART

[0003] A semiconductor device is manufactured by performing a process of transferring a pattern formed on a photomask onto a semiconductor wafer by using a lithography process and an etching process, and repeating the process. In the manufacturing process of a semiconductor device, the quality of the lithography process and the etching process, the occurrence of foreign matter, and the like greatly affect the yield of manufactured semiconductor devices. Therefore, it is important to detect the occurrence of abnormality or defect in the manufacturing process at an early stage or in advance. For this reason, in the manufacturing process of a semiconductor device, measurement and inspection of a pattern formed on a semiconductor wafer are performed. In particular, with the recent progress of miniaturization and the three-dimensionalization of semiconductor devices in recent years, it has become increasingly important to accurately manage the superposition of patterns between different processes.

[0004] In the conventional device, the position of a pattern created in each process has been measured based on reflected light obtained by irradiating a semiconductor device with light, and the amount of overlay deviation of the patterns between different processes has been measured. However, with the progress of miniaturization of patterns, it is difficult to obtain required detection accuracy by a method for detecting a deviation amount with light. Therefore, there is an increasing need to measure the amount of overlay deviation of the patterns using a scanning electron microscope having a higher resolution than light.

[0005] For example, PTL 1 proposes a method for measuring a deviation amount between two patterns formed by double patterning using a scanning electron microscope. In PTL 1, two patterns formed on the surface of a semiconductor device become measurement targets. Therefore, the scanning electron microscope can easily obtain those images.

[0006] Further, the scanning electron microscope disclosed in PTL 2 detects a signal generated from a pattern formed on a surface in an irradiation region with a secondary electron detector and

detects a signal generated from a pattern on a lower layer in the irradiation region with a backscattered electron detector. Based on two signals detected at the same time, the amount of overlay deviation between the surface pattern and the lower layer pattern is measured. In general, secondary electrons contain more information on the sample surface, and backscattered electrons contain more information on the inside the lower layer than that on the sample surface.

[0007] However, when the amount of overlay deviation between the surface pattern and the lower layer pattern is measured using the device disclosed in PTL 2, the following problem occurs. That is, in the device of PTL 2, the pattern measurement positions in both images are determined based on either the pattern position of the surface pattern in the secondary electron image or the pattern position of the lower layer pattern in the backscattered electron image. In this case, if the positional deviation between the surface pattern and the lower layer pattern is large, the pattern that is not used for pattern alignment may not be able to be measured normally due to a deviation in the measurement position. Further, when the surface pattern and the lower layer pattern are repetitive patterns, there is a concern that a measurement value shifted by the repetition period may be calculated if the overlay deviation between the surface pattern and the lower layer pattern becomes large.

CITATION LIST

Patent Literature

[0008] PTL 1: JP-A-2010-85138 [0009] PTL 2: JP-A-2014-86393

SUMMARY OF INVENTION

Technical Problem

[0010] The present invention has been made in consideration of the above technical problems, and an object thereof is to provide a charged particle beam device that can stably perform correct measurement of the amount of overlay deviation even when the amount of overlay deviation between patterns located in different layers is large.

Solution to Problem

[0011] The charged particle beam device of the present invention includes a charged particle beam irradiation unit that irradiates a sample with a charged particle beam, a first detection unit that detects secondary electrons from the sample, a second detection unit that detects backscattered electrons from the sample, and an image processing unit that generates a first image including an image of a first pattern located on the surface of the sample based on an output of the first detection unit, and generates a second image including an image of a second pattern located in a lower layer than the surface of the sample based on an output of the second detection unit. A control unit adjusts the position of a measurement area in the first image based on a first template image for the first image, and adjusts the position of a measurement area in the second image based on a second template image for the second image.

Advantageous Effects of Invention

[0012] According to the present invention, it is possible to provide a charged particle beam device that can perform stable and correct measurement even when the amount of overlay deviation between patterns located in different layers is large.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a schematic diagram showing a schematic configuration of a scanning electron microscope (SEM) as a charged particle beam device according to a first embodiment.

[0014] FIG. 2 is a schematic diagram schematically showing the operation of an image processing unit 19.

[0015] FIG. 3 is a flowchart illustrating a procedure of overlay measurement of a wafer (sample) in

the scanning electron microscope according to the first embodiment.

[0016] FIGS. **4A** and **4B** are schematic diagrams showing a configuration example of a wafer **11** as a sample.

[0017] FIGS. **5A** to **5F** are schematic diagrams illustrating a specific example of a method for registering a template image and a measurement area in a recipe.

[0018] FIGS. **6A** and **6B** are schematic diagrams illustrating details of the procedure of steps **S34c** to **34d** in the flowchart of FIG. **3**.

[0019] FIGS. **7A** to **7C** are schematic diagrams illustrating details of the procedure of steps **S34c** to **34d** in the flowchart of FIG. **3**.

[0020] FIGS. **8A** to **8D** are schematic diagrams illustrating details of the procedure of steps **S34c** to **34d** in the flowchart of FIG. **3**.

[0021] FIGS. **9A** to **9D** are schematic diagrams illustrating a specific example of a calculation process performed by a control unit **20**.

[0022] FIGS. **10A** and **10B** are schematic diagrams showing a configuration example of the wafer **11** to be measured in a second embodiment.

[0023] FIG. **11** illustrates an example of a graphical user interface (GUI) screen **111** of a scanning electron microscope (SEM) according to the second embodiment.

[0024] FIG. **12** illustrates an example of the graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment.

[0025] FIG. **13** illustrates an example of the graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment.

[0026] FIG. **14** illustrates an example of the graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment.

[0027] FIG. **15** illustrates an example of the graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment.

[0028] FIG. **16** illustrates an example of the graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment.

[0029] FIGS. **17A** and **17B** are flowcharts illustrating a procedure of overlay deviation measurement in the device according to the second embodiment.

[0030] FIGS. **18A** to **18D** are schematic diagrams showing an example of a configuration of the wafer **11** to be measured in a device according to a third embodiment.

[0031] FIGS. **19A** to **19C** are schematic diagrams illustrating a method for registering a template image according to a comparative example.

[0032] FIGS. **20A** to **20C** are schematic diagrams illustrating a method for registering a template image according to the third embodiment.

[0033] FIG. **21** is an example of the GUI screen **111** for registering a template image in the device according to the third embodiment.

[0034] FIG. **22** is an example of the GUI screen **111** for registering a template image in the device according to the third embodiment.

[0035] FIGS. **23A** to **23C** are schematic diagrams showing an example of a configuration of the wafer **11** to be measured in a fourth embodiment.

[0036] FIGS. **24A** to **24F** are schematic diagrams illustrating a procedure for determining an offset amount of a measurement point with reference to offset data of a nearby pattern position.

DESCRIPTION OF EMBODIMENTS

[0037] Hereinafter, the present embodiment will be described with reference to the accompanying drawings. In the accompanying drawings, functionally the same elements may be represented by the same numbers. Although the accompanying drawings show embodiments and implementation examples in accordance with the principles of the present disclosure, they are for the understanding of the present disclosure, and are not intended to be used for interpreting the present disclosure in a limited manner. The description in this specification is merely exemplary, and is not intended to

limit the scope of the claims or the application of the present disclosure in any way.

[0038] Although the present embodiment is described in sufficient detail for those skilled in the art to implement the present disclosure, it is necessary to understand that other implementations and forms are possible, and the change in configurations or structures and the substitution of various elements can be made without departing from the scope and spirit of the technical idea of the present disclosure. Therefore, the following description should not be construed as being limited thereto.

First Embodiment

[0039] First, a schematic configuration of a scanning electron microscope (SEM) as a charged particle beam device according to a first embodiment will be described with reference to FIG. 1. This SEM includes a column **1** which is an electron optical system, and a sample chamber **2**.

[0040] The column **1** includes an electron gun **3** for generating an electron beam (charged particle beam) to be irradiated, a condenser lens **4**, an aligner **5**, an ExB filter **6**, a deflector **7**, and an objective lens **8**, and functions as a charged particle beam irradiation unit. The condenser lens **4** and the objective lens **8** focus the electron beam generated by the electron gun **3** and irradiate a wafer **11** as a sample with the electron beam. The deflector **7** deflects the electron beam according to the applied voltage in order to scan the electron beam on the wafer **11**. The aligner **5** is configured to generate an electric field for aligning the electron beam with respect to the objective lens **8**. The ExB filter **6** is a filter for taking in the secondary electrons emitted from the wafer **11** into a secondary electron detector **9**.

[0041] Further, the column **1** and the sample chamber **2** are provided with the secondary electron detector **9** (first detection unit) for detecting secondary electrons from the wafer **11** (sample) and a backscattered electron detector **10** (second detection unit) for detecting backscattered electrons from the wafer **11**. The wafer **11** is placed on an XY stage **13** installed in the sample chamber **2**. On the XY stage **13**, a standard sample for beam calibration can be placed in addition to the wafer **11**. Further, an optical microscope **14** for optically observing the wafer **11** is provided above the XY stage **13** in order to align the wafer **11**.

[0042] The SEM further includes amplifiers **15** and **16**, an electron optical system controller **17**, a stage controller **18**, an image processing unit **19**, and a control unit **20**. The amplifiers **15** and **16** amplify the detection signals from the secondary electron detector **9** and the backscattered electron detector **10** and output the amplified detection signals to the image processing unit **19**. The electron optical system controller **17** controls the aligner **5**, the ExB filter **6**, the deflector **7**, and the like in the column **1** according to a control signal from the control unit **20**.

[0043] The stage controller **18** outputs a drive signal for driving the XY stage **13** according to a control signal from the control unit **20**. The control unit **20** can be configured by, for example, a general-purpose computer.

[0044] According to the amplified detection signals received from the amplifiers **15** and **16**, the image processing unit **19** generates an image P1 (first image P1) of the surface of the wafer **11** obtained based on the secondary electrons, and an image P2 (second image P2) of a lower layer of the surface obtained based on the backscattered electrons. The image processing unit **19** includes, for example, an edge extraction unit **1901**, a smoothing processing unit **1902**, and a matching processing unit **1903**.

[0045] The edge extraction unit **1901** extracts the edges of the pattern included in the image (first image P1) of the surface of the wafer **11** and the image (second image P2) of the lower layer than the surface. The smoothing processing unit **1902** executes a smoothing process on the first image P1 and the second image P2, for example, as preprocessing of edge extraction by the edge extraction unit **1901**.

[0046] FIG. 2 schematically shows the operation of the image processing unit **19**. As schematically shown, the matching processing unit **1903** performs matching between the first image P1 after the smoothing process and the edge extraction, and a template image T1 for the first image P1. Further,

the matching processing unit **1903** performs matching between the second image P2 after the smoothing process and the edge extraction, and a template image T2 for the second image P2. In accordance with the result of the matching, the control unit **20** adjusts the position of a measurement area and measures the amount of overlay deviation between the wafer surface and the lower layer. Here, the presence or absence of the smoothing process and the strength thereof, and whether to execute the edge extracting process can be selected for each image.

[0047] The control unit **20** controls the entire scanning electron microscope (SEM) via the electron optical system controller **17** and the stage controller **18**. Although not shown, the control unit **20** may include an input unit such as a mouse or a keyboard for a user to input instructions, a display unit for displaying a captured image or the like, and a storage unit such as a hard disk or a memory. Further, the control unit **20** may include, for example, a template image generation unit **2001** that generates the above-described template image, a deviation amount measurement unit **2002** that measures the deviation amount of the pattern in the wafer **11**, and a measurement area position adjusting unit **2003** that adjusts the position of a measurement area.

(Operation)

[0048] Next, a procedure of overlay measurement of a wafer (sample) in the scanning electron microscope according to the first embodiment will be described with reference to a flowchart of FIG. 3.

[0049] First, the control unit **20** receives a selection of the wafer **11** to be measured from a GUI screen displayed on an input unit (not shown) or a display unit (not shown), a selection of a recipe to be applied, and a selection of the necessity of performing overlay deviation measurement, and starts the overlay deviation measurement (Step S31). The control unit **20** loads the selected wafer **11** onto the XY stage **13** in the sample chamber **2** (Step S32).

[0050] Next, the control unit **20** executes the alignment of the wafer **11** based on the optical microscope image captured by the optical microscope **14** and the SEM image captured by the secondary electron detector **9** and/or the backscattered electron detector **10** (Step S33). Thereafter, the control unit **20** controls the XY stage **13** to move the wafer **11** to an overlay deviation measurement point registered in the recipe (Step S34a). The control unit **20** obtains a low-magnification SEM image to determine a position where the image should be obtained, and performs necessary adjustment (Step S34b).

[0051] In a succeeding step S34c, SEM images (the first image P1 and the second image P2) are obtained according to the imaging conditions registered in the recipe. The control unit **20** performs matching between the obtained SEM images (the first image P1 and the second image P2) and the template images T1 and T2 stored in the recipe, and adjusts the position of the measurement area in the first image P1 and the second image P2 according to the result of the matching (Step S34d). The recipe includes the first template image T1 relating to the secondary electron image (first image P1) based on a signal from the secondary electron detector **9**, and the second template image T2 relating to the backscattered electron image (second image P2) based on a signal from the backscattered electron detector **10**. In step S34c, matching between the first template image T1 and the first image P1 is performed, and matching between the second template image T2 and the second image P2 is performed. The method for generating and registering the first template image T1 and the second template image T2 will be described later. In a succeeding step S34e, a difference in the position of the pattern image between the first image P1 and the second image P2 is specified based on the result of the matching (position adjustment of the measurement area) in step S34d. Based on this difference, the amount of overlay deviation between the first image P1 and the second image P2 is calculated. The control unit **20** determines whether the calculation of the amount of overlay deviation has been completed for all the measurement points specified in the recipe (Step S34f). When the measurement points remain (No), the control unit **20** moves the irradiation range of the electron beam to the next measurement point, and repeats the acquisition of the image and the calculation of the amount of overlay deviation. When the measurement has been

completed for all the measurement points (Yes), the control unit **20** unloads the wafer **11** from the XY stage **13** (Step S35) and outputs the measurement result of the amount of overlay deviation (Step S36).

[0052] Next, with reference to FIGS. **4** and **5**, the details of the procedure for registering the recipe (template image (T1 and T2), measurement area, etc.) used in step S34c in the flowchart of FIG. **3** will be described. FIG. **4** shows a configuration example of the wafer **11** as a sample. FIG. **4(a)** is a plan view of the wafer **11**, and FIG. **4(b)** is a cross-sectional view taken along the line C-C' of FIG. **4(a)**.

[0053] In FIG. **4**, patterns **41a**, **41b**, and **41c** are patterns formed on the surface of the wafer **11**, and patterns **42a** and **42b** are patterns formed at positions in a lower layer than the surface of the wafer **11**. The patterns **41a** to **41c** on the surface are mainly imaged as the first image P1 based on the output of the secondary electron detector **9**. The lower layer patterns **42a** and **42b** are mainly imaged as the second image P2 based on the output of the backscattered electron detector **10**. In this example, it is assumed that the three centers of gravity of the patterns **41a**, **41b**, and **41c** are designed to coincide with the two centers of gravity of the patterns **42a** and **42b**.

[0054] Next, a specific example of a method for registering a template image and a measurement area in a recipe will be described with reference to FIG. **5**.

[0055] FIG. **5(a)** is a secondary electron image (first image P1) of the wafer **11** shown in FIG. **4**, and the first image P1 includes patterns **41a** to **41c** mainly formed on the surface of the wafer **11**. The first image P1 as shown in FIG. **5(a)** is displayed on a GUI screen of a display unit (not shown). When a template area **51** is set on the GUI screen by the user on the GUI screen, the control unit **20** calculates a center coordinate **52** (coordinate values (S.sub.X, S.sub.Y)) of the template area **51**, and a template image **53** (first template image T1) as shown in FIG. **5(b)** is stored as a part of the recipe.

[0056] In addition, as shown in FIG. **5(c)**, in a state where the first image P1 is displayed on the GUI screen, measurement areas **54a**, **54b**, and **54c** are set by the user at the positions of the patterns **41a** to **41c**. The measurement areas **54a** to **54c** are set as measurement areas for measuring the line width and the like of the patterns **41a** to **41c**. When the measurement areas **54a** to **54c** are set, the control unit **20** calculates center coordinates **55a** (coordinate values (M.sub.X1, M.sub.Y1)), **55b** (coordinate values (M.sub.X2, M.sub.Y2)), and **55c** (coordinate values (M.sub.X3, M.sub.Y3)) of the measurement areas **54a** to **54c**, and further stores offset amounts ($\Delta M.sub.X1$, $\Delta M.sub.Y1$), ($\Delta M.sub.X2$, $\Delta M.sub.Y2$), and ($\Delta M.sub.X3$, $\Delta M.sub.Y3$) of the center coordinates **55a** to **55c** from the center coordinate **52** in the recipe. The offset amounts of the center coordinates **55a** to **55c** are also stored in the recipe as position information of the measurement areas **54a** to **54c**. The offset amounts ($\Delta M.sub.X1$, $\Delta M.sub.Y1$), ($\Delta M.sub.X2$, $\Delta M.sub.Y2$), and ($\Delta M.sub.X3$, $\Delta M.sub.Y3$) can be expressed as follows.

$$\begin{aligned} [00001] & (M_{X1}, M_{Y1}) = (M_{X1}, M_{Y1}) - (S_X, S_Y) \\ & (M_{X2}, M_{Y2}) = (M_{X2}, M_{Y2}) - (S_X, S_Y) \\ & (M_{X3}, M_{Y3}) = (M_{X3}, M_{Y3}) - (S_X, S_Y) \end{aligned}$$

[0057] FIG. **5(d)** is a backscattered electron image (second image P2) of the wafer **11** shown in FIG. **4**, and the second image P2 includes patterns **42a** to **42b** mainly located in the lower layer than the surface of the wafer **11**. The second image P2 as shown in FIG. **5(d)** is displayed on the GUI screen. When a template area **56** is set by the user on this GUI screen, the control unit **20** calculates a center coordinate **57** (T.sub.X, T.sub.Y) of the template area **56** and stores a template image **58** (second template image T2) shown in FIG. **5(e)** as a part of the recipe.

[0058] Further, as shown in FIG. **5(f)**, in a state where the second image P2 is displayed on the GUI screen, measurement areas **59a** and **59b** are set by the user at the positions of the patterns **42a** and **42b**. The measurement areas **59a** and **59b** are set as measurement areas for measuring the line width and the like of the patterns **42a** and **42b**. When the measurement areas **59a** and **59b** are set, the control unit **20** calculates center coordinates **60a** (coordinate values (N.sub.X1, N.sub.Y1)) and **60b** (coordinate values (N.sub.X2, N.sub.Y2)) of the measurement areas **59a** and **59b**, and further

stores offset amounts ($\Delta N_{\text{sub.X1}}$, $\Delta N_{\text{sub.Y1}}$) and ($\Delta N_{\text{sub.X2}}$, $\Delta N_{\text{sub.Y2}}$) of the center coordinates **60a** and **60b** from the center coordinate **57** in the recipe. The offset amounts of the center coordinates **60a** and **60b** are stored in the recipe as position information of the measurement areas **59a** and **59b**. The offset amounts ($\Delta N_{\text{sub.X1}}$, $\Delta N_{\text{sub.Y1}}$) and ($\Delta N_{\text{sub.X2}}$, $\Delta N_{\text{sub.Y2}}$) can be expressed as follows.

$$[00002](N_{X1}, N_{Y1}) = (N_{X1}, N_{Y1}) - (T_X, T_Y) \quad (N_{X2}, N_{Y2}) = (N_{X2}, N_{Y2}) - (T_X, T_Y)$$

[0059] Next, with reference to FIGS. **6** to **8**, the details of the procedure of steps **S34c** to **34d** in the flowchart of FIG. **3** will be described. Here, as shown in FIG. **5**, a case where a deviation occurs between the first image **P1** which is a pattern on the surface of the wafer **11** and the second image **P2** in the lower layer than the surface will be described as an example.

[0060] FIG. **6(a)** is a plan view of the wafer **11**, and FIG. **6(b)** shows a cross section taken along the line D-D' of FIG. **6(a)**. Patterns **61a**, **61b**, and **61c** are patterns formed on the surface of the wafer **11**, and patterns **62a** and **62b** are patterns formed in a layer lower than the surface of the wafer **11**. In the normal wafer **11**, as shown in FIG. **4**, the patterns **42a** and **42b** are located at intermediate positions between the patterns **41a** to **41c**. On the other hand, in the example of FIG. **6**, the patterns **62a** and **62b** are formed at positions shifted in the +X direction from the positions where the patterns should be with respect to the patterns **61a**, **61b**, and **61c** on the surface. In the present embodiment, by using the two of the template image **T1** for the first image **P1** and the template image **T2** for the second image **P2**, even if an overlay deviation occurs between different layers, it is possible to accurately adjust the measurement area and to perform a stable and correct measurement.

[0061] As a comparative example, a case where only the template image **T1** for the first image **P1** is used as the template image, and the measurement position of the second image **P2** is adjusted based on the template image **T1** will be described. FIG. **7(a)** shows the first image **P1** (secondary electron image) obtained in step **S34c**, and specifies a matching range **71** in which the feature matches that of the template image **T1** stored in the recipe. Then, a deviation amount ($\Delta S_{\text{sub.X}}$, $\Delta S_{\text{sub.Y}}$) of a center coordinate **72** (coordinate values ($S_{\text{sub.X'}}$, $S_{\text{sub.Y'}}$)) of the matching range **71** from the center coordinate **52** (coordinate values ($S_{\text{sub.X}}$, $S_{\text{sub.Y}}$)) of the template area **51** is calculated.

[0062] Based on the deviation amount ($\Delta S_{\text{sub.X}}$, $\Delta S_{\text{sub.Y}}$), the coordinate values ($M_{\text{sub.X1'}}$, $M_{\text{sub.Y1'}}$), ($M_{\text{sub.X2'}}$, $M_{\text{sub.Y2'}}$), and ($M_{\text{sub.X3'}}$, $M_{\text{sub.Y3'}}$) of the center coordinates **74a**, **74b**, and **74c** of measurement areas **73a** to **73c** are as follows (See FIG. **7(b)**).

$$[00003](M'_{X1}, M'_{Y1}) = (S_X, S_Y) + (M_{X1}, M_{Y1}) + (S_X, S_Y)$$

$$(M'_{X2}, M'_{Y2}) = (S_X, S_Y) + (M_{X2}, M_{Y2}) + (S_X, S_Y)$$

$$(M'_{X3}, M'_{Y3}) = (S_X, S_Y) + (M_{X3}, M_{Y3}) + (S_X, S_Y)$$

[0063] Here, when the adjustment of the image acquisition position (process **S34b**) is performed on the first image **P1**, it is considered that there is almost no positional deviation ($\Delta S_{\text{sub.X}}$, $\Delta S_{\text{sub.Y}}$) of the surface pattern. Therefore, if $\Delta S_{\text{sub.X}}=0$ and $\Delta S_{\text{sub.Y}}=0$, the coordinate values ($M_{\text{sub.X1'}}$, $M_{\text{sub.Y1'}}$), ($M_{\text{sub.X2'}}$, $M_{\text{sub.Y2'}}$), and ($M_{\text{sub.X3'}}$, $M_{\text{sub.Y3'}}$) of the center coordinates **74a**, **74b**, and **74c** of the measurement areas **73a** to **73c** are as follows.

$$[00004](M'_{X1}, M'_{Y1}) = (S_X, S_Y) + (M_{X1}, M_{Y1}) \quad (M'_{X2}, M'_{Y2}) = (S_X, S_Y) + (M_{X2}, M_{Y2})$$

$$(M'_{X3}, M'_{Y3}) = (S_X, S_Y) + (M_{X3}, M_{Y3})$$

[0064] That is, the positions of the measurement areas **73a**, **73b**, and **73c** are the same as the positions of the measurement areas set as the recipe.

[0065] Since the adjustment of the image acquisition position is performed on the first image **P1** which is the secondary electron image, the positions of the surface patterns **61a**, **61b**, and **61c** in the image are also the same as those at the time of setting the recipe. Therefore, the respective measurement areas **73a**, **73b**, and **73c** are set correctly for the surface patterns **61a**, **61b**, and **61c**.

[0066] On the other hand, when only the template image **T1** for the first image **P1** is used as the template image and the position of the measurement area of the second image **P2** is adjusted based

on the template image T1, there is a concern that the measurement area is not set correctly for the patterns 62a and 62b, and as a result, the measurement is not performed correctly, as described later. This will be described with reference to FIG. 7(c).

[0067] FIG. 7(c) shows an example of the second image P2 (backscattered electron image) obtained simultaneously with the first image P1 (secondary electron image) of FIG. 7(a) when the wafer 11 as shown in FIG. 6 is imaged. In the second image P2, the patterns 62a and 62b are imaged shifted from the normal position in the +X direction.

[0068] In such a second image P2, center coordinates 77a and 77b (coordinate values (N.sub.X1', N.sub.Y1'), and (N.sub.X2', N.sub.Y2')) of measurement areas 76a and 76b for measuring the patterns 62a and 62b are as follows, respectively:

$$[00005](N'_{X1}, N'_{Y1}) = (S_X, S_Y) + (N_{X1}, N_{Y1}) + (S_X, S_Y)$$

$$(N'_{X2}, N'_{Y2}) = (S_X, S_Y) + (N_{X2}, N_{Y2}) + (S_X, S_Y)$$

[0069] Assuming that $\Delta S.sub.X=0$ and $\Delta S.sub.Y=0$ as described above, the following is obtained.

$$[00006](N'_{X1}, N'_{Y1}) = (S_X, S_Y) + (N_{X1}, N_{Y1})(N'_{X2}, N'_{Y2}) = (S'_X, S'_Y) + (N_{X2}, N_{Y2})$$

[0070] That is, when the position adjustment is performed based only on the template image T1 for the first image P1, the positions of the measurement areas 76a and 76b are the same as those at the time of setting the recipe. On the other hand, the patterns 62a and 62b in the second image P2 are shifted in the +X direction from the image at the time of setting the recipe. For this reason, a problem arises in that the measurement areas 76a and 76b are set to be shifted leftward from internal patterns 78a and 78b, and correct positional deviation measurement cannot be performed.

[0071] Next, referring to FIG. 8, a procedure for adjusting the position of the measurement area by using two template images of the template image T1 for the first image P1 and the template image T2 for the second image P2 according to the first embodiment of the present invention is described. FIG. 8(a) is an example of the first image P1 (secondary electron image) obtained in step 34c (FIG. 3) which is the image acquisition sequence of the recipe. Here, the matching range 71 in which the feature matches that of the template image T1 stored in the recipe is specified. Then, the deviation amount ($\Delta S.sub.X$, $\Delta S.sub.Y$) of the center coordinate 72 (coordinate values (S.sub.X', S.sub.Y')) of the matching range 71 from the center coordinate 52 (coordinate values (S.sub.X, S.sub.Y)) of the template area 51 is calculated. Assuming that $\Delta S.sub.X=0$ and $\Delta S.sub.Y=0$ for the same reason as in the example in FIG. 7, the coordinate values (M.sub.X1', M.sub.Y1'), (M.sub.X2', M.sub.Y2'), and (M.sub.X3', M.sub.Y3') of the center coordinates 74a, 74b, and 74c of the measurement areas 73a, 73b, and 73c are as follows.

$$[00007](M'_{X1}, M'_{Y1}) = (S_X, S_Y) + (M_{X1}, M_{Y1})(M'_{X2}, M'_{Y2}) = (S_X, S_Y) + (M_{X2}, M_{Y2})$$

$$(M'_{X3}, M'_{Y3}) = (S_X, S_Y) + (M_{X3}, M_{Y3})$$

[0072] Since the adjustment of the image acquisition position is performed on the first image P1 which is the secondary electron image, the positions of the surface patterns 61a, 61b, and 61c in the image are also the same as those at the time of setting the recipe. Therefore, the respective measurement areas 73a, 73b, and 73c are set correctly for the surface patterns 61a, 61b, and 61c.

[0073] FIG. 8(c) shows an example of the second image P2 (backscattered electron image) obtained simultaneously with the first image P1 (secondary electron image) of FIG. 8(a) when the wafer 11 as shown in FIG. 6 is imaged. A matching range 86 in which the feature matches that of the template image T2 stored in the recipe is specified. Then, a deviation amount ($\Delta T.sub.X$, $\Delta T.sub.Y$) of a center coordinate 87 (coordinate values (T.sub.X', T.sub.Y')) of the matching range 86 from the center coordinate 57 (coordinate values (T.sub.X, T.sub.Y)) of the template area 56 at the time of registering the recipe is calculated. Coordinate values (N.sub.X1', N.sub.Y1') and (N.sub.X2', N.sub.Y2') of center coordinates 89a and 89b of measurement areas 88a and 88b are calculated as follows based on the deviation amount of the position of the template image T2.

$$[00008](N'_{X1}, N'_{Y1}) = (T_X, T_Y) + (N_{X1}, N_{Y1}) + (T_X, T_Y)$$

$$(N'_{X2}, N'_{Y2}) = (T_X, T_Y) + (N_{X2}, N_{Y2}) + (T_X, T_Y)$$

[0074] As described above, the positions of the measurement areas **88a** and **88b** are shifted by ($\Delta T.sub.X$, $\Delta T.sub.Y$) from the position at the time of setting the recipe. ($\Delta T.sub.X$, $\Delta T.sub.Y$) indicate the deviation amount of the patterns **62a** and **62b** on the second image P2 from the position at the time of setting the recipe. Therefore, the respective measurement areas **88a** and **88b** are set at positions matching patterns **62a** and **62b**.

[0075] Next, a specific example of the calculation process performed by the control unit **20** will be described. FIG. **9(a)** is an example of the first image P1 based on the detection signal output from the secondary electron detector **9**. The first image P1 includes images of the patterns **61a**, **61b**, and **61c** formed on the surface of the wafer **11**. The change curve of the signal intensity along the line A-A' in FIG. **9(a)** has peaks at positions corresponding to the edges of the patterns **61a** to **61c** as shown in FIG. **9(b)**. The control unit **20** calculates the edge positions and the center positions of gravity of the patterns **61a** to **61c** in the measurement areas **73a**, **73b**, and **73c** according to the change in the signal intensity.

[0076] Specifically, it is assumed that the X coordinate of the edge position of the pattern **61a** in the measurement area **73a** is determined to be a.sub.1 and a.sub.2; the X coordinate of the edge position of the pattern **61b** in the measurement area **73b** is determined to be a.sub.3 and a.sub.4; and the X coordinate of the edge position of the pattern **61c** in the measurement area **73c** is determined to be a.sub.5 and a.sub.6. In this case, the X coordinates a.sub.c1, a.sub.c2, and a.sub.c3 of the center positions of gravity of the patterns **61a**, **61b**, and **61c** are given by the following equations.

$$[00009] a_{c1} = (a_1 + a_2) / 2, a_{c2} = (a_3 + a_4) / 2, a_{c3} = (a_5 + a_6) / 2$$

[0077] The X coordinate ac of the center position of gravity of the three patterns **61a** to **61c** is given by the following equation.

$$[00010] a_c = (a_{c1} + a_{c2} + a_{c3}) / 3$$

[0078] FIG. **9(c)** is an example of the second image P2 based on the detection signal output from the backscattered electron detector **10**. The second image P2 includes images of the patterns **62a** and **62b** formed in the lower layer than the surface of the wafer **11**. As shown in FIG. **9(d)**, in the change curve of the signal intensity along the line B-B' in FIG. **9(c)**, the signal intensity changes depending on the difference of the substances constituting the patterns in the portions of the patterns **62a** and **62b**. That is, the second image P2 is an image having high contrast in the portions of the patterns **62a** and **62b**. The control unit **20** calculates the edge position and the center positions of gravity of the patterns **62a** and **62b** in the measurement areas **88a** and **88b** according to the change of the contrast, that is, the change of the signal intensity.

[0079] Specifically, it is assumed that the X coordinate of the edge position of the pattern **62a** in the measurement area **88a** is determined to be b.sub.1 and b.sub.2, and the X coordinate of the edge position of the pattern **62b** in the measurement area **88b** is determined to be b.sub.3 and b.sub.4. In this case, the X coordinates b.sub.c1 and b.sub.c2 of the center positions of gravity of the patterns **62a** and **62b** are given by the following equations.

$$[00011] b_{c1} = (b_1 + b_2) / 2, b_{c2} = (b_3 + b_4) / 2$$

[0080] Further, the X coordinate b.sub.c of the center position of gravity of the two patterns **62a** and **62b** is given by the following equation.

$$[00012] b_c = (b_{c1} + b_{c2}) / 2$$

[0081] Then, the amount of overlay deviation Δx between the patterns **61a** to **61c** on the surface and the patterns **62a** to **62b** in the lower layer thereof is obtained by the following equation.

$$[00013] \Delta x = b_c - a_c$$

[0082] As described above, according to the first embodiment, the first image P1 including the image of the pattern located on the surface of the wafer **11** is generated based on the output of the secondary electron detector **9**, and the second image P2 including the image of the pattern located in the lower layer than the surface of the wafer **11** is generated based on the output of the backscattered electron detector **10**. Then, for the first image P1, the measurement area is adjusted

by the template image T1 generated for the first image P1, and for the second image P2, the measurement area is adjusted by the template image T2 separately generated for the second image P2. As a result, the amount of overlay deviation is calculated. Therefore, according to the first embodiment, even when the amount of overlay deviation of patterns formed over a plurality of layers on the wafer **11** is large, correct measurement of the amount of overlay deviation can be stably performed.

Second Embodiment

[0083] Next, a scanning electron microscope (SEM) as a charged particle beam device according to a second embodiment of the present invention will be described with reference to FIGS. **10** to **17**. The device of the second embodiment is similar to that of the first embodiment in that the template image T1 for a secondary electron image and the template image T2 for a backscattered electron image are used. This embodiment is different from the first embodiment in a user interface screen for registering a template image and the like.

[0084] In the second embodiment, for convenience of description, a case where a pattern is formed over three layers on the wafer **11** as shown in FIG. **10** will be described as an example. However, it is needless to say that the device of the second embodiment is not limited to the measurement of a wafer having such a structure. FIG. **10(a)** is a plan view of the wafer **11** viewed from the irradiation direction of the irradiation electrons, and FIG. **10(b)** is a cross-sectional view taken along the line E-E' of FIG. **10(a)**.

[0085] Patterns **101a** and **101b** are formed on a surface S0 of the wafer **11**, and a pattern **103** is formed on a first layer S1 below the surface S0 (inside the wafer **11**). Further, patterns **102a** and **102b** are formed in a second layer S2 lower than the first layer S1. In the design data, the center positions of gravity of the two patterns **101a** and **101b**, the center positions of gravity of the two patterns **102a** and **102b**, and the center position of gravity of the pattern **103** of the wafer **11** shown in FIG. **10** are all designed to coincide with each other.

[0086] Next, an example of a graphical user interface (GUI) screen **111** of the scanning electron microscope (SEM) according to the second embodiment will be described with reference to FIGS. **11** to **16**. On the GUI screen **111**, various setting operations including registration and deletion of template images, registration and deletion of measurement conditions, and registration and deletion of measurement coordinates can be performed. As described above, in the device according to the present embodiment, the template image T1 for the first image P1 which is a secondary electron image and the template image T2 for the second image P2 which is a backscattered electron image can be registered as a part of the data constituting the recipe. When the wafer **11** as shown in FIG. **10** is the target, template images T2' and T2'' can be registered for the second image P2 for each of the multiple layers S1 and S2.

[0087] Referring to FIGS. **11** and **12**, a procedure of registering the template image T1 for the first image P1 (secondary electron image) and registering the measurement condition on the GUI screen **111** will be described. As shown in FIG. **11**, on the GUI screen **111**, a map display area **112**, an image display area **113**, and a recipe condition display area **114** are arranged.

[0088] The map display area **112** is an area capable of selectively displaying a wafer map image enabling to grasp the entire image of the wafer **11** or a chip map image enabling to grasp the structure of a chip constituting a part of the wafer **11**. Switching between the display of the wafer map image and the display of the chip map image can be performed by clicking a wafer map selection button **115a** or a chip map selection button **115b** disposed below the map display area **112**. FIG. **11** shows a state in which the chip map is selected and displayed. In order to indicate that the chip map is selected, the chip map selection button **115b** is highlighted. By clicking a desired position (**116**) of the chip map image in a state where the chip map image is displayed in the map display area **112**, the XY stage **13** can be moved to the selected position (**116**) in the chip map image.

[0089] The image display area **113** is an area for displaying a captured SEM image (first image P1

(secondary electron image) or second image P2 (backscattered electron image)). The image display area **113** can selectively display the first image P1 or the second image P2, and the display can be switched by clicking a secondary electron image selection button **117a** or a backscattered electron image selection button **117b**. FIG. **11** exemplarily shows a state in which the secondary electron image selection button **117a** is clicked, and thereby the image display area **113** displays the first image P1. At a position adjacent to the image display area **113**, a display magnification change button **118**, an image acquisition button **119**, an image condition setting button **120**, a template selection button **121**, a measurement area selection button **122**, and a measurement point selection button **148** are displayed.

[0090] The display magnification change button **118** is a button for changing the display magnification of the SEM image (first image P1 or second image P2) displayed in the image display area **113**. The image acquisition button **119** is a button for obtaining an SEM image and displaying the obtained image in the image display area **113**, and for giving an instruction to store the image data of the SEM image in a memory (not shown). The image condition setting button **120** is a button for setting image conditions other than the display magnification.

[0091] The template selection button **121** is a button for giving an instruction to display a screen for selecting a template image. The measurement area selection button **122** is a button for setting a mode for setting a measurement area for the first image P1 or the second image P2. The measurement point selection button **148** is a button for selecting a mode for selecting a measurement point in the first image P1 or the second image P2.

[0092] The recipe condition display area **114** is a display area for registering and deleting template images, measurement conditions, and measurement coordinates as data constituting a part of the recipe. Above the recipe condition display area **114**, a template registration tab **123**, a measurement condition registration tab **124**, and a measurement coordinate registration tab **125** are arranged.

[0093] When the template registration tab **123** is clicked, a screen for registering or deleting a template image is displayed in the recipe condition display area **114**. On the template image registration screen of the recipe condition display area **114**, as an example, display fields of template images are arranged for each type of the image. The template image display field may include, for example, a first registration screen **126** for registering the template image T1 for the first image P1 (secondary electron image of the surface S0), a second registration screen **127** for registering the template image T2' (BSE-1) for the first layer S1 of the backscattered electron image, and a third registration screen **128**(BSE-2) for registering the template image T2'' for the second layer S2 of the backscattered electron image.

[0094] When the measurement condition registration tab **124** is clicked, a screen for registering or deleting various measurement conditions is displayed in the recipe condition display area **114**(details will be described later with reference to FIGS. **12** and **14**). Further, when the measurement coordinate registration tab **125** is clicked, a screen for registering or deleting measurement coordinates is displayed in the recipe condition display area **114**(details will be described later with reference to FIG. **16**).

[0095] When the image acquisition button **119** is clicked after the XY stage **13** has moved to the position **116**, an SEM image (first image P1 and second image P2) is obtained. When registering the template image T1 for the first image P1 which is the secondary electron image, first, the secondary electron image selection button **117a** is clicked to display the first image P1 in the image display area **113**. Subsequently, a template selection mode is set by the template selection button **121**. Thereafter, the template area **126** is selected based on the first image P1 displayed in the image display area **113**.

[0096] When the register button **129b** below the template image T1 is clicked in a state where the template image T1 is displayed on the first registration screen **126**, the displayed template image T1 is registered. When the registration is completed, the registered template image **129a** is displayed on the first registration screen **126**. The registered template image **129a** can also be

deleted by clicking a delete button **129c**. The registration and deletion of the template image T2 (T2' and T2'') for the second image P2 can be performed by the same procedure using the registration screens **127** and **128**, the register buttons **130b** and **131b**, and the delete buttons **130c** and **131c**.

[0097] FIG. **12** shows a state in which the measurement condition registration tab **124** is clicked on the GUI screen **111**, and a measurement condition registration screen for the first image P1 is displayed in the recipe condition display area **114**. In the registration of the measurement conditions for the first image P1 which is the secondary electron image, the measurement area selection mode is set by clicking the measurement area selection button **122** in a state where the first image P1 is displayed in the image display area **113**. On the first image P1 displayed in the image display area **113**, the measurement areas **140a** and **140b** are selected.

[0098] When the measurement condition registration tab **124** is clicked after the measurement area selection button **122** is clicked, a measurement location display area **141**, a pattern position measurement condition display area **142**, and an overlay deviation measurement condition display area **143** are displayed in the recipe condition display area **114**.

[0099] In the measurement location display area **141**, the first image P1 or the second image P2, and a graphic (e.g., a dotted rectangular graphic) indicating the measurement area are displayed. In the pattern position measurement condition display area **142**, symbols indicating the type of the image to be measured (the first image P1 or the second image P2), symbols indicating the type of the template image to be used, and measurement conditions such as the measuring direction of the center of gravity are displayed in a list format in a pattern position measurement condition list **144a**. When registering the displayed measurement condition, a register button **144b** is clicked. When editing the displayed measurement condition, an edit button **144c** is clicked, and when deleting the measurement condition, a delete button **144d** is clicked.

[0100] The overlay deviation measurement condition display area **143** is an area for registering, editing, or deleting measurement conditions when measuring the overlay deviation of patterns formed over a plurality of layers of the wafer **11**. The registration in an overlay deviation measurement condition list **145a** is performed by displaying a window with a register button **145b**. The registration of the overlay deviation measurement condition is possible only when at least two or more measurement conditions are registered in the pattern position measurement condition list **144a**.

[0101] If the registered overlay deviation measurement condition is desired to be changed, an edit button **145c** is clicked to display a window for edition. If the overlay deviation measurement condition is desired to be deleted, a delete button **145d** is clicked to display a window for deletion.

[0102] Next, a procedure for registering the template image T2 for the second image P2 (backscattered electron image) and registering the measurement condition will be described with reference to FIGS. **13** and **14**. As in the case of the first image P1, on the GUI screen **111**, the map display area **112**, the image display area **113**, and the recipe condition display area **114** are arranged.

[0103] In FIG. **13**, when the backscattered electron image selection button **117b** is clicked, the second image P2 is displayed in the image display area **113**. In this state, when the template selection button **121** is clicked, the template selection mode is set. After that, the template area **151** is selected based on the second image P2 displayed in the image display area **113**.

[0104] When the register button **130b** is clicked in a state where the template image T2' is displayed on the second registration screen **127**, the displayed template image T2' is registered. When the registration is completed, the registered template image **130a** (T2') is displayed on the second registration screen **127**. The registered template image **130a** can be deleted by clicking the delete button **130c**.

[0105] Similarly, when a template area **152** is selected in a state where the second image P2 is displayed in the image display area **113**, and the register button **131b** of the third registration screen

128 in the recipe condition display area **114** is clicked, the displayed template image **T2''** is registered. At this time, a registered template image **131a** (**T2''**) is displayed on the third registration screen **128**, but the template image **131a** conceals patterns in different layers, and thereby a mask area **131d** to be excluded from the matching value calculation target is set. The registered template image **131a** can be deleted by clicking the delete button **131c**.

[0106] FIG. **14** shows the GUI screen **111** in which the second image **P2** is displayed in the image display area **113**, the measurement condition registration tab **124** is clicked in a state where a measurement area **161** is selected, and a screen for registering measurement conditions for the second image **P2** is displayed in the recipe condition display area **114**. In the registration of the measurement conditions for the second image **P2** for the first layer **S1**, the measurement area selection mode is set by clicking the measurement area selection button **122** in a state where the second image **P2** and the measurement area **161** are displayed in the image display area **113**. On the second image **P2** displayed in the image display area **113**, the measurement area **161** matching the pattern **103** is selected.

[0107] When the measurement condition registration tab **124** is clicked after the measurement area selection button **122** is clicked, the measurement location display area **141**, the pattern position measurement condition display area **142**, and the overlay deviation measurement condition display area **143** are displayed in the recipe condition display area **114**. Hereinafter, various measurement conditions are registered in the same manner as in the case of the first image **P1**.

[0108] FIG. **15** shows the GUI screen **111** in which the second image **P2** is displayed in the image display area **113**, the measurement condition registration tab **124** is clicked in a state where measurement areas **171a** and **171b** are selected, and a screen for registering measurement conditions for the second image **P2** is displayed in the recipe condition display area **114**. In the registration of the measurement conditions for the second image **P2** for the second layer **S2**, the measurement area selection mode is set by clicking the measurement area selection button **122** in a state where the second image **P2** and the measurement areas **171a** and **171b** are displayed in the image display area **113**. On the second image **P2** displayed in the image display area **113**, the measurement areas **171a** and **171b** matching the patterns **102a** and **102b** are selected.

[0109] When the register button **144b** is clicked in a state where the measurement areas **171a** and **171b** are selected, the measurement condition of the second image **P2** for the second layer **S2** is registered in the third row of the pattern position measurement condition list. Here, in the measurement location display area **141**, the measurement conditions selected in the pattern position measurement condition list **144a** are graphically displayed.

[0110] The registration of the overlay deviation measurement condition can be performed by selecting a reference measurement condition (Base), measurement conditions for calculating the deviation from the reference, and the measuring direction of the center of gravity (X direction, Y direction, and XY direction) in the overlay deviation measurement condition list **145a** displayed by the register button **145b** of the overlay deviation measurement condition area **143**. Here, as the measuring direction of the center of gravity, only the direction measured under the measurement condition can be selected. The registered overlay deviation measurement condition can be changed by the edit button **145c**, and the condition can be deleted by the delete button **145d**.

[0111] FIG. **16** shows the GUI screen **111** in which the measurement coordinate selection tab **125** is clicked in a state where the first image **P1** is displayed in the image display area **113**, and a screen for registering measurement coordinates is displayed in the recipe condition display area **114**.

[0112] As shown in FIG. **16**, when a selection point **146** on the map is clicked in a state where the wafer map is displayed in the map display area **112**, the movement to any point on the wafer **11** is possible. In the registration of the measurement coordinates, the XY stage **13** is moved to a position where the measurement coordinates are desired to be registered while using the wafer map and the chip map properly. When the measurement point selection mode is set by clicking the measurement point selection button **148**, a measurement point selection mark **147** is displayed at the center

portion of the image display area **113**. In this state, when the measurement coordinate registration tab **125** is clicked, a screen for registering measurement coordinates as shown in FIG. **16** is displayed in the recipe condition display area **114**. The screen for registering measurement coordinates includes, for example, a measurement coordinate table **150a**, a register button **150b**, an edit button **150c**, and a delete button **150d**.

[0113] The measurement coordinate table **150a** includes a list of the coordinates of the registered measurement points. When register button **150b** is clicked, the coordinates of measurement point selection mark **147** indicated in the image display area **113** are registered in the measurement coordinate table **150**. The measurement coordinates can be finely adjusted by the edit button **150c**, and unnecessary measurement coordinates can be deleted by the delete button **150d**.

[0114] Hereinafter, a procedure of the overlay deviation measurement in the device according to the second embodiment will be described with reference to the flowchart in FIG. **17**. FIG. **17(a)** is a flowchart illustrating the overall flow of the procedure, and FIG. **17(b)** is a flowchart illustrating step **S184** in more detail.

[0115] First, for all the registered template images (for example, **T1**, **T2'**, and **T2''**), the offset amount from the template image in the obtained SEM image is calculated by matching between the template image and the obtained SEM image (the first image **P1** or the second image **P2**) (Steps **S182** and **S183**).

[0116] Next, the center position of gravity of the pattern is measured for all the measurement conditions registered in the recipe (Steps **S184** and **S185**). Here, the center position of gravity of the pattern can be measured by performing correction of the measurement position based on the offset amount of the corresponding template image (Step **S184a**), detection of the pattern edge position (Step **S184b**), and calculation of the center position of gravity based on the pattern edge position (Step **S184c**) in this order. Thereafter, the measurement area is adjusted based on the template image registered as a recipe, and the amount of overlay deviation between the patterns is calculated according to the adjustment amount (Steps **S186** and **S187**).

Third Embodiment

[0117] Next, a scanning electron microscope (SEM) as a charged particle beam device according to a third embodiment of the present invention will be described with reference to FIGS. **18** to **22**. The device according to the third embodiment is similar to that of the above-described embodiments in that the template image **T1** for the secondary electron image (first image **P1**) and the template image **T2** for the backscattered electron image (second image **P2**) are used. This embodiment is different from the above-described embodiments in the method for registering a template image.

[0118] FIG. **18** shows an example of the configuration of the wafer **11** to be measured in the device according to the third embodiment. FIG. **18(a)** is a front view of the wafer **11** viewed from the incident direction of the irradiation electrons, and FIG. **18(b)** shows a cross section taken along the line F-F' of FIG. **18(a)**.

[0119] A pattern **201** having a vertical direction as a longitudinal direction is formed on the surface of the wafer **11**, and patterns **202a** and **202b** having a lateral direction as a longitudinal direction are formed in a lower layer of the surface. The patterns **202a** and **202b** intersect the pattern **201** with the direction intersecting the pattern **201** as a longitudinal direction.

[0120] FIG. **18(c)** is a secondary electron image (first image **P1**) of the wafer **11**, which includes an image of the pattern **201** formed on the surface. FIG. **18(d)** is a backscattered electron image (second image **P2**) of the wafer **11**, in which a part of the patterns **202a** and **202b** in the lower layer of the wafer **11** is overlapped with the pattern **201** and is hidden.

[0121] A method for registering a template image according to a comparative example will be described with reference to FIG. **19**. In this comparative example, one template area is selected so as to include the entire lower layer patterns **211a** and **211b**. Such a method may cause the following problem in a wafer having an overlapping pattern arrangement in the upper and lower layers as shown in FIG. **19**.

[0122] In the backscattered electron image (second image P2) shown in FIG. 19(a), the lower layer patterns **211a** and **211b** are formed without being overlay-deviated from the surface pattern **212**. In this case, a template area **213** is selected so as to include the entire lower layer patterns **211a** and **211b**.

[0123] FIGS. 19(b) and 19(c) show the case where lower layer patterns **214a** and **214b** are formed at positions shifted to the right with respect to a surface pattern **215**. In this case, as shown in FIG. 19(b), a template area **216** needs to be selected so as to include the lower layer patterns **214a** and **214b**.

[0124] However, when the contrast of the surface pattern **215** is high with respect to the backscattered electron image (second image P2), as shown in FIG. 19(c), a template area **217** that includes the surface pattern **215** but does not include the entire lower layer patterns **214a** and **214b** may be selected. In this case, an appropriate template image T2 cannot be set for the lower layer patterns.

[0125] Therefore, in the third embodiment, the following method is employed for registering a template image on a wafer having an overlapping pattern as shown in FIG. 18 to solve this problem.

[0126] A method for registering a template image according to the third embodiment will be described with reference to FIG. 20. When the second image P2 as shown in FIG. 20(a) is obtained, a plurality of (for example, two) template areas (a first template area **223** and a second template area **224**) can be set for lower layer patterns **221a** and **221b**. The first template area **223** is set to include ends on one side (left ends) of the patterns **221a** and **221b**. On the other hand, the second template area **224** is set to include ends on the other side (right ends) of the patterns **221a** and **221b**. The template image T2 is set according to the two types of template areas **223** and **224**.

[0127] On the other hand, as shown in FIG. 20(b), lower layer patterns **225a** and **225b** are shifted to the left with respect to a surface pattern **226**, and the right ends of the lower layer patterns **225a** and **225b** may not have a position where the template area is arranged. In this case, a template area **227** is set only at the left ends of the lower layer patterns **225a** and **225b**. When this one template area **227** is set, a corresponding template image T2 can be set.

[0128] Further, as shown in FIG. 20(c), lower layer patterns **228a** and **228b** are shifted to the right with respect to a surface pattern **229**, and the left ends of the lower layer patterns **228a** and **228b** may not have a position where the template area is arranged. In this case, a template area **230** is set only at the right ends of the lower layer patterns **228a** and **228b**. When this one template area **230** is set, a corresponding template image T2 can be set.

[0129] In this way, by allowing a plurality of template areas (the first template area and the second template area) to be set in the second image P2, even when the lower layer pattern is shifted in any direction with respect to the surface pattern, an appropriate template image can be set.

[0130] Referring to FIGS. 21 and 22, an example of the GUI screen **111** for registering a template image in the device according to the third embodiment and an operation procedure thereof will be described. FIGS. 21 and 22 describe the case where two template images are registered for one second image P2, and also describe the effects thereof.

[0131] A procedure for registering a template image for the second image P2 which is a backscattered electron image, on the GUI screen **111** will be described with reference to FIG. 21. In the screen shown in FIG. 21, the template selection button **121** is clicked to set the template selection mode. Then, template areas **231a** and **231b** are selected based on the second image P2 displayed in the image display area **113**. The template areas **231a** and **231b** are preferably selected so that the image of the surface pattern **201** is not included.

[0132] When the register button **129c** is clicked in a state where the two template areas **231a** and **231b** are selected as described above, the template image **129a** corresponding to the template areas **231a** and **231b** is registered on the first registration screen **126**.

[0133] Next, a procedure for registering another template image for the second image P2 on the

GUI screen **111** will be described with reference to FIG. **22**. In the screen of FIG. **22**, the template selection button **121** is clicked to set the template selection mode, and then template areas **232a** and **232b** are selected based on the second image P2 displayed in the image display area **113**. The template areas **232a** and **232b** are preferably selected at positions different from the template areas **231a** and **231b** and not including the image of the surface pattern **201**.

[0134] When a register button **129e** is clicked in a state where the two template areas **232a** and **232b** are selected as described above, a template image **129b** corresponding to the template areas **232a** and **232b** is registered on the first registration screen **126**. Here, the template image **129a** can be used as a main template image in the second image P2, and the template image **129b** can be used as a spare (sub) template image. The template image **129b** may be used only when a sufficient matching rate is not obtained with the template image **129a** and it is determined to be inappropriate. Alternatively, the respective matching rates of the template image **129a** and the template image **129b** are compared, and the one with a higher matching rate may be preferentially used. That is, when a plurality of template images are generated for the second image P2, the template image generation unit **2001** employs one of the plurality of template images based on the matching rate thereof with second image P2.

[0135] The template image **129a** and the template image **129b** may be simultaneously displayed on the first registration screen **126**, or may be selectively displayed on the first registration screen **126** using a scroll bar **233**. As a criterion for determining whether or not the template image **129a** is inappropriate, an average value of shift amounts in matching of the template image **129a** or the template image **129b** may be used in addition to or instead of the matching rate.

Fourth Embodiment

[0136] Next, a scanning electron microscope (SEM) as a charged particle beam device according to a fourth embodiment of the present invention will be described with reference to FIGS. **23** and **24**. The device according to the fourth embodiment is similar to that of the above-described embodiments in that the template image T1 for the secondary electron image and the template image T2 for the backscattered electron image are used. This embodiment is different from the above-described embodiments in a method for adjusting a measurement area and a method for specifying the amount of overlay deviation.

[0137] FIG. **23** shows an example of the configuration of the wafer **11** to be measured in the fourth embodiment. FIG. **23(a)** is a front view of the wafer **11** viewed from the incident direction of the irradiation electrons, and FIG. **23(b)** shows a cross section taken along the line G-G' of FIG. **23(a)**.

[0138] On the surface of the wafer **11**, patterns **301a**, **301b**, and **301c** having a vertical direction as a longitudinal direction are formed. The patterns **301a** to **301c** are periodically formed at equal intervals P in the X direction. Similarly, patterns **302a**, **302b**, **302c**, and **302d** having a vertical direction as a longitudinal direction are formed at positions in a lower layer than the surface of the wafer **11**. The patterns **302a** to **c** are periodically formed at equal intervals P, similarly to the patterns **301a** to **d** on the surface. On the design data, the patterns **301a** to **c** and **302a** to **d** are formed at equal intervals of P/2 when viewed from the front of the wafer **11**. As shown in FIG. **23(c)**, a template area **304** is set so as to include, for example, two pattern images **303a** and **303b** among a plurality of patterns of the backscattered electron image of the wafer **11**.

[0139] Next, a procedure for determining an offset amount of a measurement point with reference to offset data of a nearby pattern position will be described with reference to FIG. **24**. FIG. **24(a)** is a front view of the wafer **11** viewed from the incident direction of the irradiation electrons, and FIG. **24(b)** shows a cross section taken along the line H-H' of FIG. **24(a)**. On the design data, the wafer **11** has a structure in which patterns **301a** to **c** and **302a** to **d** are formed at equal intervals as shown in FIG. **23**. In the actually manufactured wafer **11**, lower layer patterns **312a**, **312b**, and **312c** are offset with respect to the patterns **301a**, **301b**, and **301c** by about a half (P/2) of the period P.

[0140] As described above, in some cases, it is unclear which direction of the left or right an offset

has occurred when there is a pattern that extends in parallel between the upper and lower layers and the offset occurs. For this reason, the device according to the fourth embodiment is configured to specify the direction in which the offset has occurred, to adjust the measurement area position according to the specified result, and further to measure the overlay deviation, by the method described later.

[0141] FIGS. 24(c) and 24(d) show examples of the backscattered electron image (second image P2) obtained by imaging the wafer 11 shown in FIG. 24(a) and matching positions 313 and 315 to be set. The second images P2 in FIGS. 24(c) and 24(d) show images after the position adjustment has been performed so that the positions of the surface patterns 301a to d coincide with the positions in the template image T1.

[0142] In the matching of the lower layer patterns 302a to d, since the lower layer patterns 302a to d are periodic, a case where the matching position 313 is selected for the lower layer patterns 312a and 312b on the left side of the image as shown in FIG. 24(c), and a case where the matching position 315 is selected for the lower layer patterns 312b and 312c on the right side of the image as shown in FIG. 24(d) can be considered. In FIG. 24(c), an arrow 314 indicates the offset amount of the lower layer pattern when the matching position 313 is selected, and an arrow 316 indicates the offset amount of the lower layer pattern when the matching position 315 is selected. The arrow 314 indicates that the lower layer pattern is offset to the left, and the arrow 316 indicates that the lower layer pattern is offset to the right.

[0143] A method for determining the amount and direction of the offset will be described with reference to FIGS. 24(e) and 24(f). As shown in FIG. 24(e), when a measurement point 317 is measured on the current wafer 11 and the offset amount and the offset direction at the measurement point 317 are determined, the offset amount at the measurement point 317 can be determined with reference to the offset amount and the offset direction at a measurement point 318 closest to the measurement point 317 where the measurement is completed.

[0144] For example, when the second image P2 and a matching position 320 as shown in FIG. 24(f) are obtained at the measurement point 317 and the offset amount and direction as shown in FIG. 24(c) are obtained at the nearest point 318, with reference to this offset amount and direction, the offset amount and direction at the measurement point 317 can be determined. When a plurality of offset amounts and offset amount candidates are obtained at the measurement point 317, a candidate that approximates the offset amount and direction at the measurement point 318 can be selected as the offset amount and direction at the measurement point 317.

[0145] In the fourth embodiment, the specification is such that the result of the measurement point closest to the current measurement point is referred to, but an approximate expression from the results of a plurality of measurement points may be used. Further, the specification may be such that the pattern offset amount at each measurement point calculated by template matching is output with the overlay deviation measurement result.

[0146] The present invention is not limited to the above-described embodiments, and includes a variety of modification examples. For example, the above-described embodiments have been described in detail for easy understanding of the present invention, and are not necessarily limited to those having all the configurations described above. Further, a part of the configuration of one embodiment can be replaced with the configuration of another embodiment, and the configuration of one embodiment can be added to the configuration of another embodiment. Also, for a part of the configuration of each embodiment, addition, deletion, or replacement of another configuration can be made. Further, each of the above-described configurations, functions, processing units, processing means, and the like may be partially or entirely realized by hardware, for example, by designing an integrated circuit.

REFERENCE SIGNS LIST

[0147] 1: column [0148] 2: sample chamber [0149] 3: electron gun [0150] 4: condenser lens [0151] 5: aligner [0152] 6: ExB filter [0153] 7: deflector [0154] 8: objective lens [0155] 9: secondary

electron detector [0156] **10**: backscattered electron detector [0157] **11**: wafer [0158] **12**: standard sample [0159] **13**: XY stage [0160] **14**: optical microscope [0161] **15**, **16**: amplifier [0162] **17**: electron optical system controller [0163] **18**: stage controller [0164] **19**: image processing unit [0165] **20**: control unit

Claims

1. A system including one or more processors, the system being configured to calculate an amount of overlay deviation between patterns from image data obtained by irradiating a sample with a charged particle beam, the one or more processors being configured to: determine, based on first image data including image data of a first pattern located on a surface of the sample, a first position of the first pattern; determine, based on second image data including image data of a second pattern located below the surface of the sample, a second position of the second pattern; adjust a position of first measurement area in the first image data based on the determined first position; adjust a position of the second measurement area in the second image data based on the determined second position; and calculate an amount of overlay deviation between the first and second patterns according to a result of position adjustment of the first and second measurement areas.
2. The system according to claim 1, wherein the first image data is secondary electron image data obtained by irradiating the sample with the charged particle beam, the second image data is backscattered electron image data obtained by irradiating the sample with the charged particle beam.
3. The system according to claim 1, wherein the one or more processors determine, from the first image data, and based on a first template, the first position of the first pattern, and determine, from the second image data, and based on a second template, the second position of the second pattern.
4. The system according to claim 1, wherein the one or more processors determine a first measurement area center position as a center position of the first measurement area, and determine a second measurement area center position as a center position of the second measurement area.
5. The system according to claim 1, wherein the first image data includes image data of a plurality of first repetition patterns as the first pattern, and the second image data includes image data of a plurality of second repetition patterns as the second pattern.
6. The system according to claim 4, wherein the one or more processors determine the first measurement area center position based on selection of the first measurement area, and determine the second measurement area center position based on selection of the second measurement area.
7. The system according to claim 6, wherein the one or more processors determine first measurement area center position as a center position of the first measurement area, and determine second measurement area center position as a center position of the second measurement area.
8. The system according to claim 7, wherein the one or more processors determine a first offset amount or a first offset direction based on the first template area center position and the first measurement area center position, or determine a second offset amount or a second offset direction based on the second template area center position and the second measurement area center position.
9. The system according to claim 8, wherein the one or more processors output or display the amount of overlay deviation, and output or display at least one of the first offset amount, the first offset direction, the second offset amount, and the second offset direction.
10. The system according to claim 1, wherein the one or more processors generate one or more first templates corresponding to the first pattern based on the first image data and generate one or second templates corresponding to the second pattern based on the second image data.
11. The system according to claim 10, wherein when the second image data includes multiple layers of image data, the one or more processors generate one or more second templates for each of the multiple layers.
12. The system according to claim 10, wherein the one or more processors exclude, conceal or

mask, from the second image data, image data of a pattern on another layer different from a layer on which the second pattern is present, to generate one or more second templates.

13. The system according to claim 10, wherein the one or more processors are configured to generate, from the second image data, a plurality of second templates that differ in area from each other.

14. The system according to claim 13, wherein the one or more processors are configured to select either one of the plurality of second template images according to a match rate with the second image data.

15. The system according to claim 5, wherein the first measurement area comprises a plurality of first repetitive measurement areas corresponding to the plurality of first repetitive patterns, and wherein the second measurement area comprises a plurality of second repetitive measurement areas corresponding to the plurality of second repetitive patterns.

16. The system according to claim 15, wherein the one or more processors determine a plurality of first repetitive measurement area center positions as center positions of the plurality of first repetitive measurement areas, and determine a plurality of second repetitive measurement area center positions as center positions of the plurality of second repetitive measurement areas.

17. A method for calculating an amount of overlay deviation between patterns from image data obtained by irradiating a sample with a charged particle beam, the method comprising: determining, based on first image data including image data of a first pattern located on a surface of the sample, a first position of the first pattern; determining, based on second image data including image data of a second pattern located below the surface of the sample, a second position of the second pattern; adjusting the position of the first measurement area in the first image data based on the determined first position; adjusting a position of the second measurement area in the second image data based on the determined second position; and calculating an amount of overlay deviation between the first pattern and the second pattern according to a result of position adjustment of the first and second measurement areas.

18. A non-transitory computer readable medium containing a computer program for calculating an amount of overlay deviation between patterns from image data obtained by irradiating a sample with a charged particle beam, the computer program being configured to perform: determining, based on first image data including image data of the first pattern located on a surface of the sample, a first position of the first pattern; determining, based on second image data including image data of the second pattern located below the surface of the sample, a second position of the second pattern; adjusting a position of the first measurement area in the first image data based on the determined first position; adjusting a position of the second measurement area in the second image data based on the determined second position; and calculating an amount of overlay deviation between the first and second patterns according to a result of position adjustment of the first and second measurement areas.

19. A system including one or more processors, the system being configured to calculate an amount of overlay deviation between patterns from image data obtained by irradiating a sample with a charged particle beam, the one or more processors being configured to: determine, based on first image data including image data of the plurality of first patterns located on a surface of the sample, positions of the plurality of first patterns; determine, based on second image data including image data of the plurality of second patterns located below the surface of the sample, positions of the plurality of second patterns; adjust positions of the plurality of measurement areas in the first image data and the second image data based on the determined positions of the plurality of first patterns and the plurality of second patterns; and calculate an amount of overlay deviation between the plurality of first patterns and the plurality of second patterns according to a result of position adjustment of the plurality of measurement areas.
